Cyber Defense Club: SSH and Networking

Oct 24 session - what we covered (full write-up in the session guide; section numbers match)

#	Topic	Summary
1	VirtIO GPU display	Switch Display from Spice to VirtIO GPU; takes effect after VM restart; viewed with Virt Viewer
2	ip command basics	`ip address`; 127.0.0.1/8 loopback; /24 = first three octets are the network, last is the host
3	The /24 -> /0 mistake	How the GUI ended up with /0, why the host then thinks everything is local, ping test with 0 replies
4	Correct network config	Your VM address, which is 144.76.90., mask 255.255.255.0, gateway 144.76.90.254, DNS 199.100.16.100; verify with ping
5	wget and missing CSS	`wget` the Google page, open via `file://` in Firefox, why it's unstyled
6	What SSH is	Bare `ssh` shows the client is preinstalled; secure shell from one computer to another
7	Enable the service	`systemctl status ssh`, `enable`, `start`, re-check: enabled + active
8	Install on a second VM	`apt update` first (and why), then `apt install openssh-server`, enable and start
9	First connection + host key	Fingerprint prompt, saved to `~/.ssh/known_hosts`, changed hash later = warning
10	Port 22 lookup	`grep 22 /etc/services` shows ssh = remote login
11	Add a user	Creating accounts in Settings > Users
12	Connecting as a user	`ssh username@<server IP>`; `hostname` and `ip address` to confirm which VM you're on
13	Users, passwd, shadow, permissions	Default home directory, account isolation, `/etc/passwd` fields and the x placeholder, `/etc/shadow` hashes and `!` service accounts, `ls -l` permission triplets
14	Next week: sshd_config	SSH vs sshd (daemon), sshd_config.d, why moving off port 22 doesn't hide the service; coming up: ListenAddress, host keys, authorized_keys, TOTP